BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Mid Term
Duration: 1 hour 15 minutes

Semester: Summer 2022
Full Marks: 25

CSE 340: Computer Architecture

Answer the following questions.
Figures in the right margin indicate marks.
Understanding the question is part of the exam.

Name: [redacted] ID: [redacted] Section: 07

1. CO1 a) **Discover** the new Execution time and SPECRatio for the Go and Hmmer program. Also, **calculate** the geometric mean of the SPECRatios. **5**

Name of Benchmark Program	Instruction Count ($\times 10^9$)	CPI	Clock cycle time(seconds $\times 10^{-9}$)	Execution Time (seconds)	Reference Time (seconds)	SPECRatio
Go	1974	1.30	476		101490	
Hmmer	8086	1.60	180		154560	

b) **Explain** how the compiler and the algorithm affect the performance. **2**

c) Is MIPS a RISC or CISC architecture? **List** the differences between these two architectures based on what we have learned about MIPS. **3**

2. CO2 a) **Write down** the equivalent MIPS code of the following code, given i is in \$s3, the base addresses of X and Y are in \$t0 and \$t1 respectively. **7**

```
C[] = concat(X[], Y[]);

concat(char A[], char B[]) {
     for (int i =0; i < 10; i = i + 1) {
          A[i] = A[i] + B[i];
     }
     return A[]
}
```

b) Consider the instruction `beq $t0, $t1, L1` 3
where the PC has the value (in hex) 0x40001000. If the offset value for L1 (in decimal) is 2, then **calculate** the address in PC

i) if \$t0 is equal to \$t1
ii) if \$t0 is not equal to \$t1

(Your answer should show all the steps associated with the calculation and the final result should be in hex format.)

3. CO1 a) Consider A = 5ED8 and B = 3C91 as signed 16-bit hexadecimal numbers. **Determine** whether A - B causes overflow or not. Show all the necessary steps. 2

b) **Calculate** the product of the two hexadecimal unsigned 4-bit integers ~~B~~ C and A using the long-multiplication approach (using the hardware with one 4-bit register and two 8-bit registers). Consider ~~B~~ C as the multiplicand and A as the multiplier in the calculation. You have to show the contents of each register on each step in an iteration. 3

Hint: The values of C and A in decimal are 12 and 10 respectively.